



Chathuni Karunarathne

Software Engineering Intern

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About me

Information Technology & Management undergraduate at the University of Moratuwa, specializing in full-stack development and scalable systems design using Java, TypeScript, Next.js, and React. Passionate about combining robust backend logic powered by PostgreSQL, Prisma, and Drizzle ORM with intuitive, high-end UI/UX prototyping. Balances academic principles with hands-on experience in requirements engineering, IoT frameworks, and AI-assisted development tools to build highly optimized, end-to-end digital solutions.

Education

BSc. (Hons) Information Technology & Management,

2024 – Present

Faculty of IT, University of Moratuwa

- **CGPA:** 3.74 / 4.00
- **Academic Honors:** Included in the Dean's List for Semester 02

G.C.E. Advanced Level, Devi Balika Vidyalaya, Colombo 08

2020 – 2023

- 2022(2023) - 3A passes in Commerce Stream (English Medium)

Technical Skills

Languages: Java, Typescript, JavaScript, C, C++, SQL, Python

Frameworks & Libraries: Tailwind CSS, Next.js, React, Node.js, Spring Boot, REST APIs

Databases & ORMs: MySQL, PostgreSQL, Drizzle, Neon, Spring Data JPA, Hibernate

DevOps Tools: Git, Github, Docker, Vercel, Expo

Hardware Platforms: ESP32, Arduino, PCB Design (EasyEDA)

Projects

Enterprise IT Asset Management System (EITAMS)

11/2025 – 09/2026

Full-Stack Developer

- Developed a scalable, enterprise-level asset registry using Next.js and React, incorporating advanced data grids, real-time updates, and automated PDF-based asset tag generation.
- Engineered relational database schemas with Drizzle ORM (PostgreSQL/MySQL) to support secure asset management, bulk transfer operations, reporting, and role-based authentication with NextAuth.
- Enhanced application maintainability by restructuring large UI components into reusable React hooks and modular components while contributing within an Agile development team.
- **Tech Stack:** Next.js, React, TypeScript, Tailwind CSS, PostgreSQL, Drizzle ORM, Git, GitHub.

PantryPal – Smart Inventory & Order Management System

09/2026 – Present

- Developing a smart inventory and order management system for small-scale cookery and bakery businesses to manage ingredients, products, recipes, and stock efficiently.
- Implementing ingredient tracking with Pantry, Refrigerator, and Freezer storage classification, along with receipt, barcode, and manual inventory entry methods.
- Building recipe-based inventory automation to automatically calculate and deduct ingredient quantities when customer orders are processed.
- Designing an AI-powered **Sunday Stock Clearance** workflow to identify ingredients requiring priority usage and support the creation of special products to reduce food waste.
- **Tech Stack:** Next.js, TypeScript, Tailwind CSS, Spring Boot, Java, Spring Data JPA, Hibernate, PostgreSQL, REST API, Git, GitHub.

- Designed a custom EasyEDA PCB for the timing and scoring system, implementing efficient seven-segment display driving using 74HC595 shift registers and BC547 transistor drivers.
- Developed the paddle control mechanism using rotary encoders, enabling accurate real-time player input for the physical Pong game.
- Implemented the countdown timing system and score display logic using multiple seven-segment displays, ensuring reliable game-state management and synchronized visual feedback.
- **Tech Stack:** EasyEDA, Arduino IDE, C++, ESP32, Arduino Mega, 74HC595, BC547 Transistors, Rotary Encoders, Seven-Segment Displays, Git, GitHub.

- Developed a Java-based desktop camping reservation management system using Swing and AWT for an intuitive graphical user interface.
- Implemented role-based authentication, MySQL database integration, and CRUD operations for managing campsites, reservations, customers, and bookings using JDBC.
- Built user-friendly interfaces with search functionality, report generation, file handling, PDF generation, and internationalization (i18n) support.
- **Tech Stack:** Java, Swing, AWT, MySQL, JDBC, i18n, File I/O, PDF Generation, Git, GitHub.

Certificates

BCS Level 4 Certificate in IT	Java Programming Certification
	SLIIT

Experience

Iot & Embedded Systems Labs (IES Labs), <i>Software Developer</i>	09/2025 – Present
• Electronic Team Member	
ICTBUS Institute, <i>Academic Instructor</i>	01/2025 – 01/2026

Community and Leadership

- Hackelite 2.0,** *IEEE WIE Student Branch Affinity Group of UOM*
Co-Chairperson
- Annual General Meeting 2025,** *IEEE Professional Communication Society*
Leading Moderator
- Road to Legacy 2.0,** *IIEE of USJ*
Lead- Delegates Handling Committee
- Binara Padura 2.0,** *Rotaract Club of University of Moratuwa*
Member - Finance Committee